Analytical Chemistry
Cumulative Exam
January 20, 2017

1. The authors provide a definition for the activity of any solute as: $a_x \equiv exp\left(\frac{G_x - G_X^o}{RT}\right)$

a) In lay person terms, explain what this definition means.

b) What does it mean for a solution to be departed from ideal behavior?

c) Give examples of how non-ideality of a solution affects different aspects of equilibrium chemistry.

2. Figure 1 from the assigned paper shows Flood's diagram for strong acids and strong bases relative to weak acids and bases.

a) Define the difference between a strong acid/weak acid and a strong base/weak base. Give examples of each.

b) The diagram shows that if you take equimolar concentrations of a strong acid vs. weak acids of varying pK_a values, the resulting pH will change. In fact, as the pK_a of the acid is increased, the change in pH is decreased. Give a mechanistic reason for why the pH shift is smaller as the pK_a of the acid increases. If it helps, consider a formal or initial concentration of 1 mM for any strong acid vs. a weak acid with pK_a values of 5,7 and 9.

3. The Debye-Hückel equation $\log \gamma = -0.50z^2\sqrt{I}$, where *I* is the ionic strength, provides an easy back of the envelope method for calculating possible changes from ideality for a solution.

a) Using the definition for K_a that includes activity coefficients, derive an equation that would relate pK_a of any acid to the solution ionic strength. If it helps, use acetic acid, $pK_a = 4.75$ as an example.

b) Based on your equation how does the pK_a of an acid shift as the ionic strength is increased?

c) Provide a chemical explanation for this shift.

4. The paper uses many terms that are common to microfluidic devices. Define the following terms.

a) Electrokinetics

b) Joule heating

c) Electrophoretic mobility

d) Brønsted-Lowry acid

e) Electroosmotic pump